

# Mathematics 2 Homework GD.2 Programming Report

## Problem 6: Comparison of Optimisation Methods

### Models and Parameters

The goal is to compare GD, Polyak GD, Nesterov GD, AdaGrad, Adam, Newton method and BFGS on the three given functions. Each method is tested from the required starting points and compared after 2, 5, 10 and 100 steps, and after 0.1, 1 and 2 seconds.

- Functions:
  - $f_1(x, y, z) = (x - z)^2 + (2y + z)^2 + (4x - 2y + z)^2 + x + y$
  - $f_2(x, y, z) = (x - 1)^2 + (y - 1)^2 + 100(y - x^2)^2 + 100(z - y^2)^2$
  - $f_3(x, y) = (1.5 - x + xy)^2 + (2.25 - x + xy^2)^2 + (2.625 - x + xy^3)^2$
- Starting points:
  - $f_1$ : (0, 0, 0) and (1, 1, 0)
  - $f_2$ : (1.2, 1.2, 1.2) and (-1, 1.2, 1.2)
  - $f_3$ : (1, 1) and (4.5, 4.5)
- GD:  $\gamma = 0.001$
- Polyak GD:  $\gamma = 0.001, \mu = 0.9$
- Nesterov GD:  $\gamma = 0.001, \mu = 0.9$
- AdaGrad:  $\gamma = 0.01, \varepsilon = 10^{-8}$
- Adam:  $\gamma = 0.001, \beta_1 = 0.9, \beta_2 = 0.999, \varepsilon = 10^{-8}$
- Newton method:  $\alpha = 1$
- BFGS: starts with the identity matrix as the inverse Hessian approximation and uses line search for the step size

### Output

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PART (a): Performance by number of steps
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Function 1) f(x,y,z) = (x-z)^2 + (2y+z)^2 + (4x-2y+z)^2 + x + y
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Starting point: [0. 0. 0.]
Method          2 steps          5 steps          10 steps          100 steps
-----
GD              -0.003947        -0.009615        -0.018469        -0.111403
Polyak          -0.005707        -0.024321        -0.065437        -0.197883
Nesterov        -0.005692        -0.024118        -0.064726        -0.197856
AdaGrad         -0.031685        -0.056334        -0.081489        -0.179946
Adam            -0.003971        -0.009836        -0.019351        -0.135067
Newton          -0.197917        -0.197917        -0.197917        -0.197917
BFGS            -0.133891        -0.197917        -0.197917        -0.197917

    2 steps: 1. Newton (-0.197917), 2. BFGS (-0.133891), 3. AdaGrad (-0.031685)
    5 steps: 1. Newton (-0.197917), 2. BFGS (-0.197917), 3. AdaGrad (-0.056334)
   10 steps: 1. Newton (-0.197917), 2. BFGS (-0.197917), 3. AdaGrad (-0.081489)
  100 steps: 1. Newton (-0.197917), 2. BFGS (-0.197917), 3. Polyak (-0.197883)

Starting point: [1. 1. 0.]
Method          2 steps          5 steps          10 steps          100 steps
```

|          |           |           |           |           |
|----------|-----------|-----------|-----------|-----------|
| GD       | 10.242712 | 9.277925  | 8.017875  | 1.936482  |
| Polyak   | 9.913269  | 7.106921  | 4.133920  | -0.197397 |
| Nesterov | 9.923807  | 7.202177  | 4.077512  | -0.196978 |
| AdaGrad  | 10.562652 | 10.185127 | 9.756435  | 6.970224  |
| Adam     | 10.948073 | 10.870466 | 10.741916 | 8.634301  |
| Newton   | -0.197917 | -0.197917 | -0.197917 | -0.197917 |
| BFGS     | 1.174951  | -0.197912 | -0.197917 | -0.197917 |

2 steps: 1. Newton (-0.197917), 2. BFGS (1.174951), 3. Polyak (9.913269)  
5 steps: 1. Newton (-0.197917), 2. BFGS (-0.197912), 3. Polyak (7.106921)  
10 steps: 1. BFGS (-0.197917), 2. Newton (-0.197917), 3. Nesterov (4.077512)  
100 steps: 1. BFGS (-0.197917), 2. Newton (-0.197917), 3. Polyak (-0.197397)

Function 2)  $f(x,y,z) = (x-1)^2 + (y-1)^2 + 100(y-x^2)^2 + 100(z-y^2)^2$

Starting point: [1.2 1.2 1.2]

| Method   | 2 steps   | 5 steps   | 10 steps | 100 steps |
|----------|-----------|-----------|----------|-----------|
| GD       | 0.042372  | 0.018223  | 0.018109 | 0.016875  |
| Polyak   | 4.697759  | 0.161202  | 4.266084 | 0.001071  |
| Nesterov | 0.489933  | 0.127005  | 3.144180 | 0.092156  |
| AdaGrad  | 8.155288  | 5.941211  | 4.056005 | 0.102337  |
| Adam     | 11.143472 | 10.481404 | 9.440934 | 1.156201  |
| Newton   | 0.035418  | 0.000041  | 0.00e+00 | 0.00e+00  |
| BFGS     | 3.720412  | 0.079308  | 0.004580 | 0.00e+00  |

2 steps: 1. Newton (0.035418), 2. GD (0.042372), 3. Nesterov (0.489933)  
5 steps: 1. Newton (0.000041), 2. GD (0.018223), 3. BFGS (0.079308)  
10 steps: 1. Newton (0.00e+00), 2. BFGS (0.004580), 3. GD (0.018109)  
100 steps: 1. Newton (0.00e+00), 2. BFGS (0.00e+00), 3. Polyak (0.001071)

Starting point: [-1. 1.2 1.2]

| Method   | 2 steps   | 5 steps   | 10 steps    | 100 steps |
|----------|-----------|-----------|-------------|-----------|
| GD       | 5.291304  | 4.242976  | 4.203009    | 4.133886  |
| Polyak   | 15.851510 | 4.754752  | 4.512499    | 3.186586  |
| Nesterov | 11.885893 | 10.586343 | 3.688114    | 1.891565  |
| AdaGrad  | 9.751536  | 7.412732  | 5.748465    | 4.257943  |
| Adam     | 13.249176 | 12.455460 | 11.223334   | 4.314755  |
| Newton   | 9.061843  | 4.236227  | 1156.427674 | 0.00e+00  |
| BFGS     | 7.583594  | 3.606559  | 3.222680    | 0.00e+00  |

2 steps: 1. GD (5.291304), 2. BFGS (7.583594), 3. Newton (9.061843)  
5 steps: 1. BFGS (3.606559), 2. Newton (4.236227), 3. GD (4.242976)  
10 steps: 1. BFGS (3.222680), 2. Nesterov (3.688114), 3. GD (4.203009)  
100 steps: 1. Newton (0.00e+00), 2. BFGS (0.00e+00), 3. Nesterov (1.891565)

Function 3)  $f(x,y) = (1.5-x+xy)^2 + (2.25-x+xy^2)^2 + (2.625-x+xy^3)^2$

Starting point: [1. 1.]

| Method   | 2 steps   | 5 steps   | 10 steps  | 100 steps |
|----------|-----------|-----------|-----------|-----------|
| GD       | 12.807585 | 11.251866 | 9.459676  | 2.061266  |
| Polyak   | 12.218716 | 8.027551  | 4.083937  | 0.086604  |
| Nesterov | 12.253178 | 8.228678  | 4.191109  | 0.083665  |
| AdaGrad  | 13.737011 | 13.321514 | 12.833988 | 9.338297  |
| Adam     | 14.147722 | 14.064791 | 13.926945 | 11.494177 |
| Newton   | 14.203125 | 14.203125 | 14.203125 | 14.203125 |
| BFGS     | 4.438122  | 0.064655  | 0.000502  | 0.00e+00  |

2 steps: 1. BFGS (4.438122), 2. Polyak (12.218716), 3. Nesterov (12.253178)  
5 steps: 1. BFGS (0.064655), 2. Polyak (8.027551), 3. Nesterov (8.228678)  
10 steps: 1. BFGS (0.000502), 2. Polyak (4.083937), 3. Nesterov (4.191109)

100 steps: 1. BFGS (0.00e+00), 2. Nesterov (0.083665), 3. Polyak (0.086604)

Starting point: [4.5 4.5]

| Method   | 2 steps                                                                                                   | 5 steps       | 10 steps      | 100 steps     |
|----------|-----------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|
| GD       | 158739770121987533884194532016112943874223603347742869817126839129809789643699299141736030738929550863892 |               |               |               |
| Polyak   | 158739770110264167766432167796168549408401906999293953550056072754411231382215405779655054027806418791301 |               |               |               |
| Nesterov | 143867079511859484589601344341991383686924399573471337081051797113266111492122667365660696638127687735115 |               |               |               |
| AdaGrad  | 169646.559640                                                                                             | 165200.723108 | 160163.711645 | 127388.016334 |
| Adam     | 174199.026683                                                                                             | 173281.246259 | 171762.026592 | 147002.272716 |
| Newton   | 15043.650877                                                                                              | 418.804026    | 14.275383     | 14.203125     |
| BFGS     | 2894.304950                                                                                               | 32.420068     | 8.462121      | 7.379384      |

2 steps: 1. BFGS (2894.304950), 2. Newton (15043.650877), 3. AdaGrad (169646.559640)

5 steps: 1. BFGS (32.420068), 2. Newton (418.804026), 3. AdaGrad (165200.723108)

10 steps: 1. BFGS (8.462121), 2. Newton (14.275383), 3. AdaGrad (160163.711645)

100 steps: 1. BFGS (7.379384), 2. Newton (14.203125), 3. AdaGrad (127388.016334)

=====  
PART (b): Performance by wall-clock time  
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Function 1)  $f(x,y,z) = (x-z)^2 + (2y+z)^2 + (4x-2y+z)^2 + x + y$

Starting point: [0. 0. 0.]

| Method | 0.1s  | 1.0s  | 2.0s  |
|--------|-------|-------|-------|
| -----  | ----- | ----- | ----- |

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:452: RuntimeWarning: overflow encountered in scalar multiply  
r3 = 2.625 - x + x\*y\*\*3

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:453: RuntimeWarning: overflow encountered in scalar power  
return r1\*\*2 + r2\*\*2 + r3\*\*2

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:463: RuntimeWarning: overflow encountered in scalar multiply  
r3 = 2.625 - x + x\*y\*\*3

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:464: RuntimeWarning: overflow encountered in scalar multiply  
df\_dx = 2\*r1\*(y-1) + 2\*r2\*(y\*\*2-1) + 2\*r3\*(y\*\*3-1)

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:465: RuntimeWarning: overflow encountered in scalar multiply  
df\_dy = 2\*r1\*x + 2\*r2\*(2\*x\*y) + 2\*r3\*(3\*x\*y\*\*2)

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:450: RuntimeWarning: invalid value encountered in scalar add  
r1 = 1.5 - x + x\*y

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:451: RuntimeWarning: invalid value encountered in scalar add  
r2 = 2.25 - x + x\*y\*\*2

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:452: RuntimeWarning: invalid value encountered in scalar add  
r3 = 2.625 - x + x\*y\*\*3

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:461: RuntimeWarning: invalid value encountered in scalar add  
r1 = 1.5 - x + x\*y

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:462: RuntimeWarning: invalid value encountered in scalar add  
r2 = 2.25 - x + x\*y\*\*2

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:463: RuntimeWarning: invalid value encountered in scalar add  
r3 = 2.625 - x + x\*y\*\*3

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:451: RuntimeWarning: overflow encountered in scalar multiply  
r2 = 2.25 - x + x\*y\*\*2

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:452: RuntimeWarning: overflow encountered in scalar power  
r3 = 2.625 - x + x\*y\*\*3

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:462: RuntimeWarning: overflow encountered in scalar multiply  
r2 = 2.25 - x + x\*y\*\*2

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:463: RuntimeWarning: overflow encountered in scalar power  
r3 = 2.625 - x + x\*y\*\*3

C:\Users\tjasa\Desktop\MAT2\Assignment35\hw2\_programming.py:464: RuntimeWarning: overflow encountered in scalar power  
df\_dx = 2\*r1\*(y-1) + 2\*r2\*(y\*\*2-1) + 2\*r3\*(y\*\*3-1)

|          |           |           |           |
|----------|-----------|-----------|-----------|
| GD       | -0.136473 | -0.197917 | -0.197917 |
| Polyak   | -0.197917 | -0.197917 | -0.197917 |
| Nesterov | -0.197917 | -0.197917 | -0.197917 |
| AdaGrad  | -0.197917 | -0.197917 | -0.197917 |
| Adam     | -0.197917 | -0.197917 | -0.197917 |
| Newton   | -0.197917 | -0.197917 | -0.197917 |

BFGS                -0.197917   -0.197917   -0.197917

0.1s: 1. Polyak (-0.197917), 2. Nesterov (-0.197917), 3. AdaGrad (-0.197917)  
1.0s: 1. GD (-0.197917), 2. Polyak (-0.197917), 3. Nesterov (-0.197917)  
2.0s: 1. GD (-0.197917), 2. Polyak (-0.197917), 3. Nesterov (-0.197917)

Starting point: [1. 1. 0.]

| Method   | 0.1s      | 1.0s      | 2.0s      |
|----------|-----------|-----------|-----------|
| GD       | -0.197917 | -0.197917 | -0.197917 |
| Polyak   | -0.197917 | -0.197917 | -0.197917 |
| Nesterov | -0.197917 | -0.197917 | -0.197917 |
| AdaGrad  | -0.197917 | -0.197917 | -0.197917 |
| Adam     | -0.197917 | -0.197917 | -0.197917 |
| Newton   | -0.197917 | -0.197917 | -0.197917 |
| BFGS     | -0.197917 | -0.197917 | -0.197917 |

0.1s: 1. GD (-0.197917), 2. Polyak (-0.197917), 3. Nesterov (-0.197917)  
1.0s: 1. GD (-0.197917), 2. Polyak (-0.197917), 3. Nesterov (-0.197917)  
2.0s: 1. GD (-0.197917), 2. Polyak (-0.197917), 3. Nesterov (-0.197917)

Function 2)  $f(x,y,z) = (x-1)^2 + (y-1)^2 + 100(y-x^2)^2 + 100(z-y^2)^2$

Starting point: [1.2 1.2 1.2]

| Method   | 0.1s     | 1.0s     | 2.0s     |
|----------|----------|----------|----------|
| GD       | 0.000001 | 1.86e-26 | 1.86e-26 |
| Polyak   | 2.47e-31 | 2.47e-31 | 2.47e-31 |
| Nesterov | 0.000033 | 0.000033 | 0.000033 |
| AdaGrad  | 0.000001 | 0.000001 | 0.000001 |
| Adam     | 8.54e-18 | 8.54e-18 | 8.54e-18 |
| Newton   | 0.00e+00 | 0.00e+00 | 0.00e+00 |
| BFGS     | 0.00e+00 | 0.00e+00 | 0.00e+00 |

0.1s: 1. Newton (0.00e+00), 2. BFGS (0.00e+00), 3. Polyak (2.47e-31)  
1.0s: 1. Newton (0.00e+00), 2. BFGS (0.00e+00), 3. Polyak (2.47e-31)  
2.0s: 1. Newton (0.00e+00), 2. BFGS (0.00e+00), 3. Polyak (2.47e-31)

Starting point: [-1. 1.2 1.2]

| Method   | 0.1s     | 1.0s     | 2.0s     |
|----------|----------|----------|----------|
| GD       | 0.000025 | 4.65e-27 | 4.65e-27 |
| Polyak   | 2.47e-31 | 2.47e-31 | 2.47e-31 |
| Nesterov | 0.000063 | 0.000063 | 0.000063 |
| AdaGrad  | 0.000018 | 0.000002 | 0.000002 |
| Adam     | 2.76e-17 | 2.08e-17 | 2.08e-17 |
| Newton   | 0.00e+00 | 0.00e+00 | 0.00e+00 |
| BFGS     | 0.00e+00 | 0.00e+00 | 0.00e+00 |

0.1s: 1. Newton (0.00e+00), 2. BFGS (0.00e+00), 3. Polyak (2.47e-31)  
1.0s: 1. Newton (0.00e+00), 2. BFGS (0.00e+00), 3. Polyak (2.47e-31)  
2.0s: 1. Newton (0.00e+00), 2. BFGS (0.00e+00), 3. Polyak (2.47e-31)

Function 3)  $f(x,y) = (1.5-x+xy)^2 + (2.25-x+xy^2)^2 + (2.625-x+xy^3)^2$

Starting point: [1. 1.]

| Method   | 0.1s      | 1.0s      | 2.0s      |
|----------|-----------|-----------|-----------|
| GD       | 0.000044  | 8.14e-26  | 8.14e-26  |
| Polyak   | 1.52e-24  | 5.55e-29  | 5.55e-29  |
| Nesterov | 9.95e-22  | 5.08e-29  | 5.08e-29  |
| AdaGrad  | 4.16e-26  | 4.16e-26  | 4.16e-26  |
| Adam     | 0.00e+00  | 0.00e+00  | 0.00e+00  |
| Newton   | 14.203125 | 14.203125 | 14.203125 |

```

BFGS          0.00e+00  0.00e+00  0.00e+00

0.1s: 1. Adam (0.00e+00),  2. BFGS (0.00e+00),  3. AdaGrad (4.16e-26)
1.0s: 1. Adam (0.00e+00),  2. BFGS (0.00e+00),  3. Nesterov (5.08e-29)
2.0s: 1. Adam (0.00e+00),  2. BFGS (0.00e+00),  3. Nesterov (5.08e-29)

Starting point: [4.5 4.5]
Method          0.1s          1.0s          2.0s
-----
GD              174813.363281  174813.363281  174813.363281
Polyak          174813.363281  174813.363281  174813.363281
Nesterov        174813.363281  174813.363281  174813.363281
AdaGrad         1.039947      1.13e-26      1.13e-26
Adam            60.488334      0.00e+00      0.00e+00
Newton          14.203125      14.203125     14.203125
BFGS            7.341767      7.341767      7.341767

0.1s: 1. AdaGrad (1.039947),  2. BFGS (7.341767),  3. Newton (14.203125)
1.0s: 1. Adam (0.00e+00),  2. AdaGrad (1.13e-26),  3. BFGS (7.341767)
2.0s: 1. Adam (0.00e+00),  2. AdaGrad (1.13e-26),  3. BFGS (7.341767)

```

## Discussion

- Newton and BFGS performed best on  $f_1$ , reaching the minimum value in very few steps.
- On  $f_2$ , Newton and BFGS reached the best final values, while first order methods converged more slowly.
- On  $f_3$  from  $(4.5, 4.5)$ , GD, Polyak GD and Nesterov GD diverged with the fixed learning rate.
- The time comparison gives the same main conclusion as the step comparison, second order methods perform best overall.

# Problem 7: Linear Regression

## Models and Parameters

The goal is to fit  $g(x) = kx + n$  to generated points  $(i, i + \nu_i)$ , where  $\nu_i \sim U[0, 1]$ . The methods are compared for  $N = 50, 100, 1000, 10000, 100000, 1000000$ .

- Model:  $g(x) = kx + n$
- Objective:  $f(k, n) = \frac{1}{N} \sum_{i=1}^N (ki + n - y_i)^2$
- Methods: GD, SGD, Newton method, BFGS and L BFGS
- Starting point:  $(k, n) = (0, 0)$
- GD:  $\gamma_{GD} = 1/\beta$ , where  $\beta = \lambda_{\max}(H)$
- SGD:  $\gamma_{SGD} = 0.01/\beta$ , where  $\beta = \lambda_{\max}(H)$
- Newton method: exact Hessian step
- BFGS: starts with the identity matrix as the inverse Hessian approximation and uses line search for the step size
- L BFGS: memory parameter  $m = 10$  and line search for the step size
- Reference optimum: computed with `np.linalg.lstsq`

## Output

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PROBLEM 7: Linear Regression  g(x) = kx + n
=====

```

N = 50 optimum = 0.074997 k\* = 0.9968 n\* = 0.6181

| Method | 2 steps   | 5 steps  | 10 steps | 100 steps |
|--------|-----------|----------|----------|-----------|
| GD     | 0.158750  | 0.158608 | 0.158373 | 0.154245  |
| SGD    | 37.415972 | 0.531921 | 0.172218 | 0.408689  |
| Newton | 0.074997  | 0.074997 | 0.074997 | 0.074997  |
| BFGS   | 0.097247  | 0.074997 | 0.074997 | 0.074997  |
| L-BFGS | 0.158750  | 0.074997 | 0.074997 | 0.074997  |

2 steps: 1. Newton (0.074997), 2. BFGS (0.097247), 3. L-BFGS (0.158750)  
5 steps: 1. Newton (0.074997), 2. BFGS (0.074997), 3. L-BFGS (0.074997)  
10 steps: 1. Newton (0.074997), 2. BFGS (0.074997), 3. L-BFGS (0.074997)  
100 steps: 1. Newton (0.074997), 2. BFGS (0.074997), 3. L-BFGS (0.074997)

N = 100 optimum = 0.070529 k\* = 0.9979 n\* = 0.5915

| Method | 2 steps    | 5 steps  | 10 steps | 100 steps |
|--------|------------|----------|----------|-----------|
| GD     | 0.152357   | 0.152321 | 0.152262 | 0.151198  |
| SGD    | 199.138136 | 2.078328 | 0.987079 | 0.867290  |
| Newton | 0.070529   | 0.070529 | 0.070529 | 0.070529  |
| BFGS   | 0.091614   | 0.070529 | 0.070529 | 0.070529  |
| L-BFGS | 0.152357   | 0.070529 | 0.070529 | 0.070529  |

2 steps: 1. Newton (0.070529), 2. BFGS (0.091614), 3. L-BFGS (0.152357)  
5 steps: 1. Newton (0.070529), 2. BFGS (0.070529), 3. L-BFGS (0.070529)  
10 steps: 1. BFGS (0.070529), 2. L-BFGS (0.070529), 3. Newton (0.070529)  
100 steps: 1. BFGS (0.070529), 2. L-BFGS (0.070529), 3. Newton (0.070529)

N = 1000 optimum = 0.084937 k\* = 1.0000 n\* = 0.4962

| Method | 2 steps      | 5 steps    | 10 steps   | 100 steps |
|--------|--------------|------------|------------|-----------|
| GD     | 0.146019     | 0.146019   | 0.146018   | 0.146010  |
| SGD    | 21120.949714 | 291.735811 | 220.153288 | 0.317829  |
| Newton | 0.084937     | 0.084937   | 0.084937   | 0.084937  |
| BFGS   | 0.100253     | 0.084937   | 0.084937   | 0.084937  |
| L-BFGS | 0.146019     | 0.146019   | 0.146018   | 0.146010  |

2 steps: 1. Newton (0.084937), 2. BFGS (0.100253), 3. L-BFGS (0.146019)  
5 steps: 1. BFGS (0.084937), 2. Newton (0.084937), 3. GD (0.146019)  
10 steps: 1. BFGS (0.084937), 2. Newton (0.084937), 3. L-BFGS (0.146018)  
100 steps: 1. BFGS (0.084937), 2. Newton (0.084937), 3. GD (0.146010)

N = 10000 optimum = 0.083062 k\* = 1.0000 n\* = 0.4955

| Method | 2 steps        | 5 steps      | 10 steps     | 100 steps |
|--------|----------------|--------------|--------------|-----------|
| GD     | 0.144390       | 0.144390     | 0.144390     | 0.144390  |
| SGD    | 2153728.793274 | 31391.501920 | 23911.516889 | 0.298709  |
| Newton | 0.083062       | 0.083062     | 0.083062     | 0.083062  |
| BFGS   | 0.098399       | 0.083062     | 0.083062     | 0.083062  |
| L-BFGS | 0.144390       | 0.144390     | 0.144390     | 0.144390  |

2 steps: 1. Newton (0.083062), 2. BFGS (0.098399), 3. GD (0.144390)  
5 steps: 1. BFGS (0.083062), 2. Newton (0.083062), 3. GD (0.144390)  
10 steps: 1. BFGS (0.083062), 2. Newton (0.083062), 3. GD (0.144390)  
100 steps: 1. BFGS (0.083062), 2. Newton (0.083062), 3. GD (0.144390)

N = 100000 optimum = 0.083243 k\* = 1.0000 n\* = 0.5008

| Method | 2 steps          | 5 steps        | 10 steps       | 100 steps |
|--------|------------------|----------------|----------------|-----------|
| GD     | 0.145948         | 0.145948       | 0.145948       | 0.145948  |
| SGD    | 215471348.673178 | 3151506.079921 | 2400957.668067 | 0.147731  |

|        |          |          |          |          |
|--------|----------|----------|----------|----------|
| Newton | 0.083243 | 0.083243 | 0.083243 | 0.083243 |
| BFGS   | 0.098920 | 0.083243 | 0.083243 | 0.083243 |
| L-BFGS | 0.145948 | 0.145948 | 0.145948 | 0.145948 |

2 steps: 1. Newton (0.083243), 2. BFGS (0.098920), 3. GD (0.145948)  
5 steps: 1. Newton (0.083243), 2. BFGS (0.083243), 3. GD (0.145948)  
10 steps: 1. BFGS (0.083243), 2. Newton (0.083243), 3. GD (0.145948)  
100 steps: 1. BFGS (0.083243), 2. Newton (0.083243), 3. GD (0.145948)

N = 1000000 optimum = 0.083310 k\* = 1.0000 n\* = 0.4997

| Method | 2 steps            | 5 steps          | 10 steps         | 100 steps |
|--------|--------------------|------------------|------------------|-----------|
| GD     | 0.145724           | 0.145724         | 0.145724         | 0.145724  |
| SGD    | 21543794981.418667 | 315203014.796151 | 240322675.063204 | 0.216176  |
| Newton | 0.083310           | 0.083310         | 0.083310         | 0.083310  |
| BFGS   | 0.098914           | 0.083310         | 0.083310         | 0.083310  |
| L-BFGS | 0.145724           | 0.145724         | 0.145724         | 0.145724  |

2 steps: 1. Newton (0.083310), 2. BFGS (0.098914), 3. GD (0.145724)  
5 steps: 1. BFGS (0.083310), 2. Newton (0.083310), 3. GD (0.145724)  
10 steps: 1. BFGS (0.083310), 2. Newton (0.083310), 3. GD (0.145724)  
100 steps: 1. BFGS (0.083310), 2. Newton (0.083310), 3. GD (0.145724)

## Discussion

- Newton reached the optimum after 1 step because the loss is quadratic and the Hessian is exact.
- BFGS reached the optimum in very few steps for all tested values of  $N$ .
- GD stayed stable, but convergence was slow for larger  $N$ .
- SGD was less stable than GD because each step uses only one random point, so its losses were larger for large  $N$ .